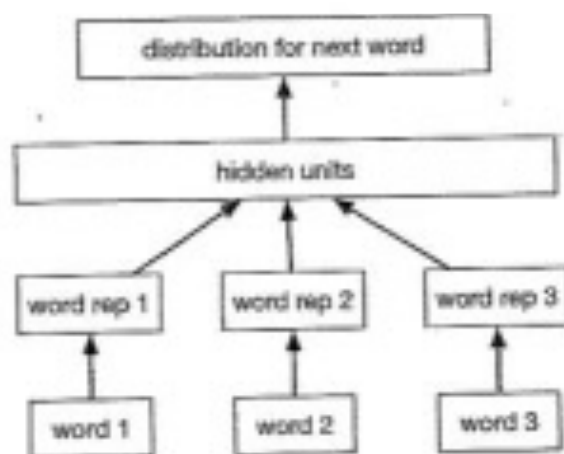


1. In autoregressive language modeling, when we generate from the model (i.e. compute samples from its distribution over sentences), the outputs feed back in to the network as inputs. What is teacher forcing, and why is it used? [8 pts]

2. Recall the neural language model architecture:



Why is it that the learned word representations for *teach* and *professor* are likely to be far apart? (It is not sufficient to say they are semantically dissimilar. Your answer should mention how the representations are used in the prediction task.) [8 pts]

3. Recall the CycleGAN model for style transfer.

(a) If the model is successful, what will it be able to do? [5 pts]

(b) What data is required to train the model? [5 pts]

⇒ (c) What might go wrong if we eliminate the discriminator terms from the cost function? [5 pts]

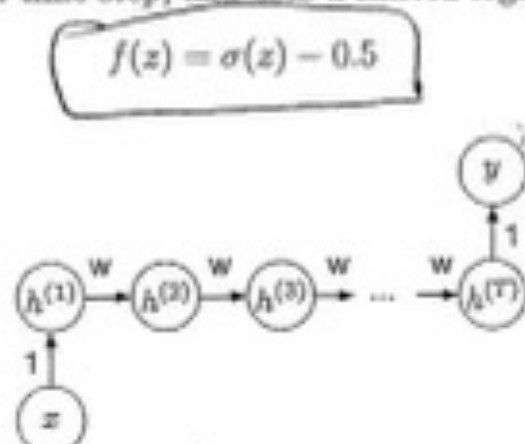
⇒ (d) What might go wrong if we eliminate the reconstruction term from the cost function? [5 pts]

4. In this course, we've discussed several cases where you want to use backprop to compute the gradient of some function with respect to the pixels of an image. Describe two of these cases. For each one, explain what function we compute the gradient of and what the gradient is useful for. (A one-sentence verbal description is sufficient; you don't need to write equations.)

(a) Case 1 [5 pts]:

(b) Case 2 [5 pts]:

5. Consider the following RNN, which has a scalar input at the first time step, makes a scalar prediction at the last time step, and uses a shifted logistic activation function:



- (a) Write the formula for the derivative  $\overline{h_t}$  as a function of  $\overline{h_{t+1}}$ , for  $t < T$ . (Use  $z_t$  to denote the input to the activation function at time  $t$ . You may write your answer in terms of  $\sigma'$ , i.e., you don't need to explicitly write out the derivative of  $\sigma$ .) [8 pts]
- (b) Suppose the input to the network is  $x = 0$ . Notice that  $\overline{h_t} = 0$  for all  $t$ . Based on your answer to part (a), determine the value  $\alpha$  such that if  $w < \alpha$ , the gradient vanishes, while if  $w > \alpha$ , the gradient explodes. You may use the fact that  $\sigma'(0) = 1/4$ . [8 pts]

## 6. Language Models and Transformers

- (a) In the encoder/decoder architecture for machine translation, the network reads a sentence and stores all the information in its hidden units. Describe two ways in which a bidirectional RNN encoder differs from this. [8 pts]

- (b) How do self-attention and cross-attention differ? [6 pts]

- (c) Describe one way in which BERT and GPT models are similar, and one way in which they differ. [8 pts]

## 7. Autoencoders

(a) What is the training objective for a standard autoencoder? [5 pts]

~~(b) How does the Variational Autoencoder differ from a standard autoencoder? [5 pts]~~

(c) Describe the two terms in the training objective of a VAE, and how they relate to the overall aim of the model. [6 pts]